COMMON CRAWL

Multilinguality at Common Crawl

Improving Language Coverage for the Largest Open Web Corpus

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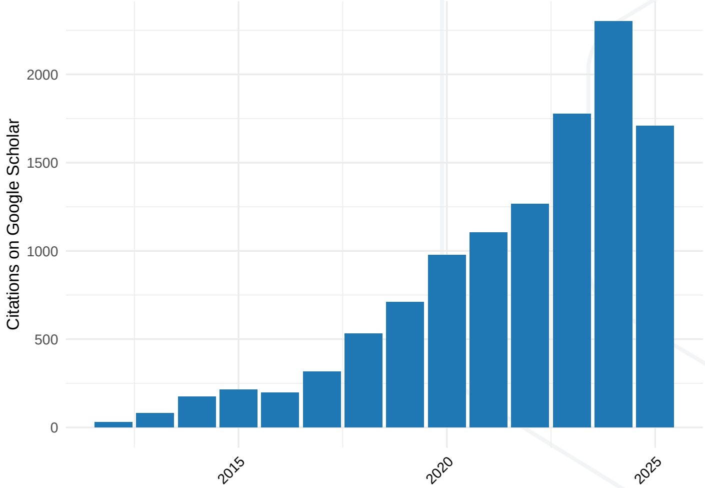
About Common Crawl

- We’re a non-profit making web data accessible to programmers and data scientists
- Free archive of the public web since 2007, founded by Gil Elbaz
- Hosted as Open Dataset on Amazon Web Services [1, 2]
- Cited in over 10,000 research papers

Papers Citing Common Crawl (through 9/25)



Common Crawl Use Cases

- Search indexes
- Sociopolitical research
- Linguistic analyses
- Internet security research
- Economic research
- AI/ML training
- A lot more...

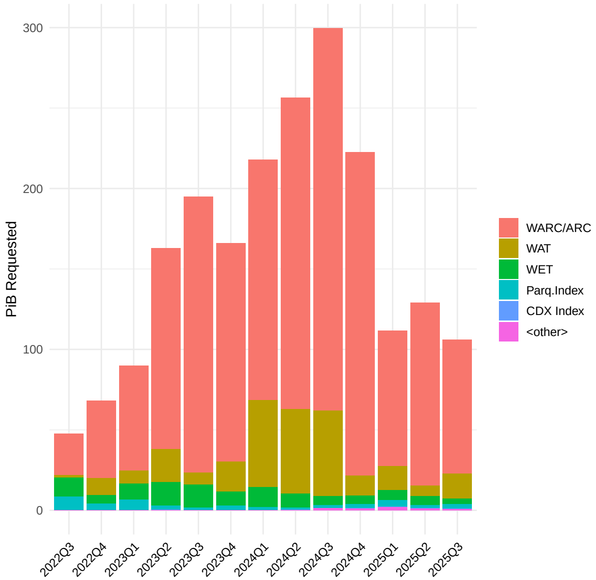
Our Mission

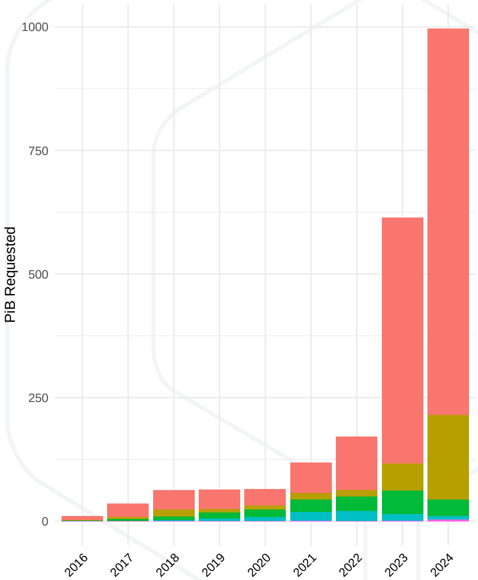
- Make high-quality public web crawl data (once limited to large search engines) available to everyone
- Support people, governments, and organizations make data-driven decisions and tackle global issues
- Build an open knowledge ecosystem that encourages sharing and innovation
- Support understanding and preservation of language, community, and culture

Data Overview

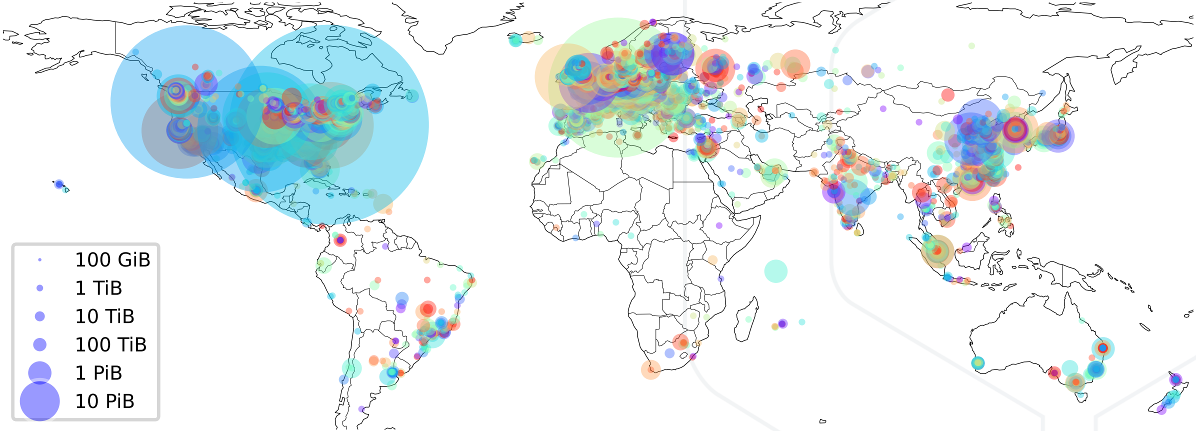
- Over 300 billion web pages spanning 17 years (2008 – 2025)
- Around 2.5 billion pages added each month
- More than 100 crawl archives released to date
- 10.1 PiB of data (Sept 2025)
- Additional data products: web graph, host index, etc.

Data Downloads





Downloads by Geographical Location



Data Collection – A Look Back In Time

Four phases of data collection using different

- Crawler implementations
- Approaches to find and sample (prioritize) seeds and URLs
- Page revisit policies

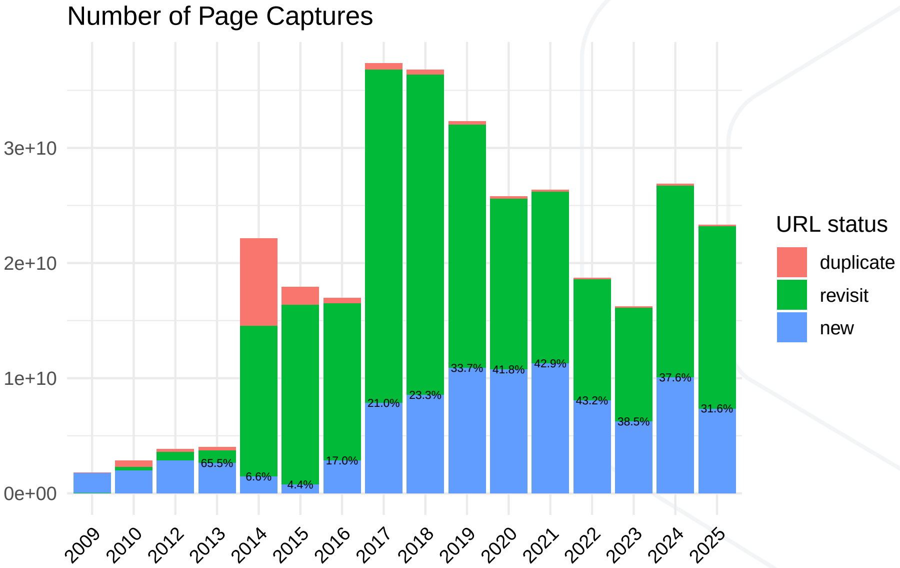
	crawler	seeds / link prioritization	revisit policy
2008–2009	Nutch	list based	
2012	in-house	page rank	
2013 – 2016	Nutch	seed donations, list based	
2017 – now	Nutch	harmonic centrality	30% new, 70% revisits based on page relevance

What Is Representative?

Aspects of representativity:

- Breadth: coverage of unique domains (web sites)
- Depth: per-site coverage
- Freshness (new content)
- Amount of (near-)duplicates (per crawl and over multiple crawls)
- Regional coverage (top-level domains, content languages)
- Content quality
- Applicable for a given data use case?

Size and Freshness



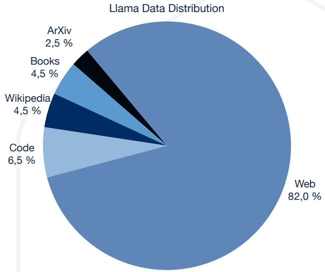
Crawler Politeness

- Crawl slowly
  - Further slow down (exponential backoff) if a site responds with errors
- Respect robots.txt rules (Robots Exclusion Protocol – [3]) and
- URI-level metatags (`<meta name=robots value=nofollow>`) [4, 5]

- CCBot identifies itself
  - User-agent string and contact information sent along with requests
  - Crawling from fixed list of IP addresses, publicly announced and verifiable via reverse DNS

Prevalence of Web Data in LLMs Today(ish)

- In practice datasets are not balanced
- Web data is always the cheapest and easiest to get
- Web data is diverse, but definitely not balanced or representative of all the language range.
- Web data always contains unwanted content (fiction, bias, propaganda).
- Programming language code (source code) is becoming ubiquitous in the data mix
- Public domain books and encyclopedic data is also common, but availability varies greatly between languages.



LLaMA Data

Source	Proportion
Web	82%
Code	6.5%
Wikipedia	4.5%
Books	4.5%
ArXiv	2.5%

Anatomy of a Data Pipeline

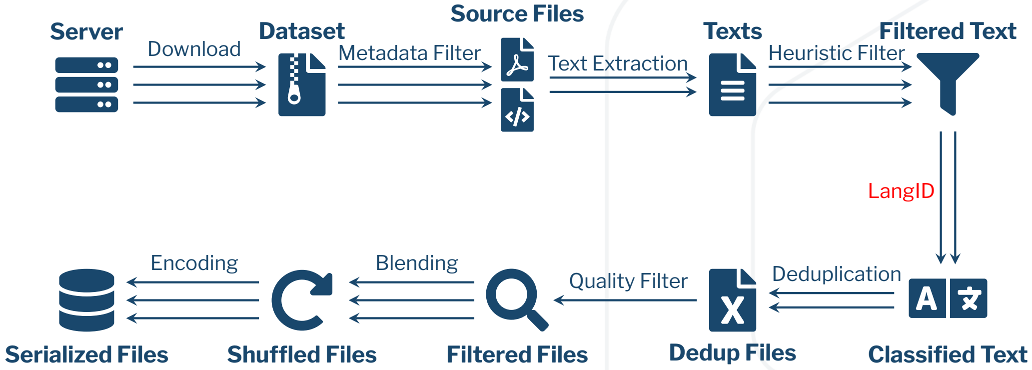


Figure 1: A diagram of a common pre-processing pipeline. Arrows represent parallel processes; the more arrows there are, the more parallelizable the specific task is.

A Survey of Pre-Processing Pipelines

- OSCAR (First use of LangID treshold for filtering) [6, 7]
- CCNet (Uses KenLM models [8] trained on Wikipedia) [9]
- mC4 (Uses word blocklists) [10]
- Refined Web (Uses Trafilatura for text extraction) [11]
- DataTrove (Uses Trafilatura and aggregates filters) [12]
- Dolma (Works on deduplication) [13]
- NeMo-Curator (Combines many heuristics and filters) [14]
- HPLT (Uses custom pipeline and has bilingual data) [15]
- MADLAD (Uses custom LangID and focuses on Multilinguality) [16]
- GlotCC (Uses Ungoliant [7] plus a new LangID model) [17]
- Community OSCAR (KenLM models trained on Adult Content) [18]
- And many others!

Multilinguality is Hard – Even with Web Data!

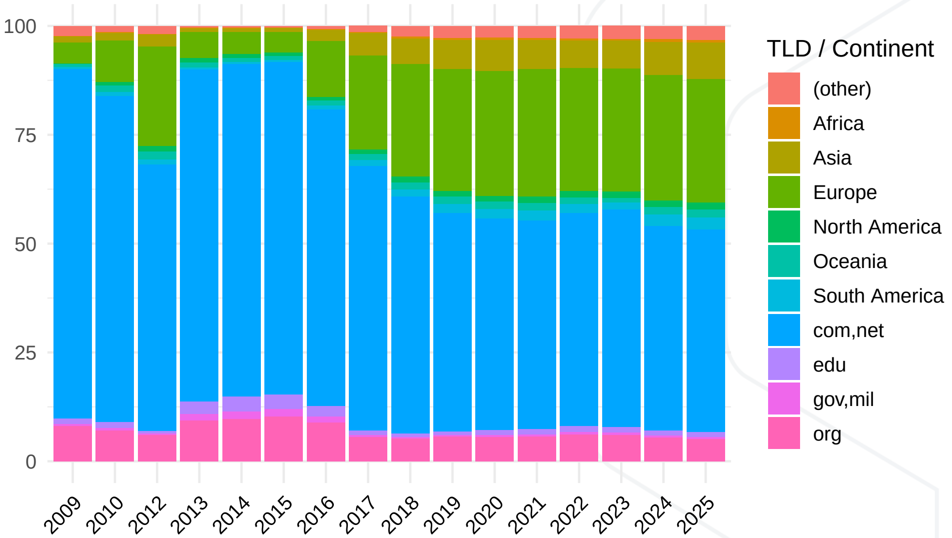
- The hope: lots of multilingual data on the web
- The reality: current pipelines struggle with multilinguality
  - **Plurality of crawl is English**
  - **Language identification isn't perfect**
  - Extraction and preprocessing not fully multilingual

Language Coverage Skews Towards English

	2018	2019	2020	2021	2022	2023	2024	2025
<other>	6.47	7.31	7.88	7.59	7.51	7.84	7.90	7.91
<unknown>	3.32	2.62	2.35	2.53	2.80	2.84	3.00	2.91
ara	0.76	0.59	0.56	0.60	0.63	0.64	0.66	0.67
ces	1.03	1.04	1.05	1.06	1.03	1.09	1.04	1.02
deu	5.15	5.47	5.57	5.63	5.60	5.79	5.36	5.57
eng	**43.96**	**43.84**	**43.20**	**44.98**	**46.64**	**45.70**	**44.40**	**44.07**
fas	0.66	0.59	0.57	0.63	0.63	0.67	0.71	0.72
fra	4.53	4.56	4.53	4.46	4.50	4.64	4.31	4.31
ind	0.75	0.74	0.76	0.83	0.75	0.82	1.01	1.09
ita	2.06	2.31	2.38	2.42	2.48	2.69	2.53	2.31
jpn	5.47	4.81	4.78	4.66	4.68	4.83	5.03	5.09
kor	0.59	0.69	0.76	0.66	0.65	0.67	0.71	0.77
nld	1.50	1.74	1.79	1.80	1.93	2.09	1.85	1.81
pol	1.68	1.70	1.68	1.62	1.62	1.70	1.80	1.75
por	1.99	2.07	2.16	2.15	1.78	1.27	2.14	2.26
rus	**9.27**	**7.41**	**7.11**	**7.07**	**5.90**	**5.84**	**6.02**	**5.97**
spa	4.18	4.16	4.25	4.33	4.36	4.58	4.56	4.46
tur	0.90	0.87	0.91	1.00	0.86	0.84	1.18	1.15
vie	0.75	0.74	0.80	0.92	0.92	1.04	1.01	1.03
zho	4.98	6.76	6.91	5.07	4.73	4.41	4.75	5.12

Percentage of yearly crawls in selected languages, as identified by CLD2 Source: [19]

Top-Level Domain Coverage Skews Western

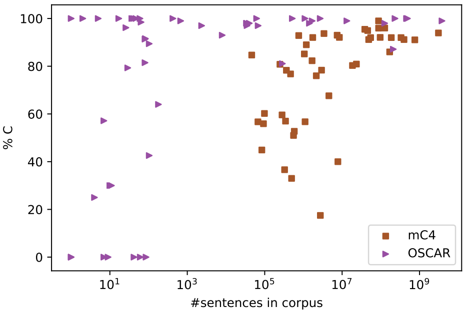


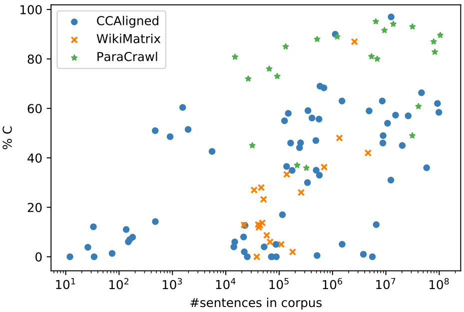
Human Evaluation Shows Big Web LID Problems

		Parallel			Monolingual	
		CCAligned	ParaCrawl v7.1	WikiMatrix	OSCAR	mC4
#langs audited / total		65 / 119	21 / 38	20 / 78	51 / 166	48 / 108
%langs audited		54.62%	55.26%	25.64%	30.72%	44.44%
#sents audited / total		8037 / 907M	2214 / 521M	1997 / 95M	3517 / 8.4B	5314 / 8.5B
%sents audited		0.00089%	0.00043%	0.00211%	0.00004%	0.00006%
macro	C	29.25%	76.14%	23.74%	87.21%	72.40%
	X	29.46%	19.17%	68.18%	-	-
	WL	9.44%	3.43%	6.08%	6.26%	15.98%
	NL	31.42%	1.13%	1.60%	6.54%	11.40%
	offensive	0.01%	0.00%	0.00%	0.14%	0.06%
	porn	5.30%	0.63%	0.00%	0.48%	0.36%
micro	C	53.52%	83.00%	50.58%	98.72%	92.66%
	X	32.25%	15.27%	47.10%	-	-
	WL	3.60%	1.04%	1.35%	0.52%	2.33%
	NL	10.53%	0.69%	0.94%	0.75%	5.01%
	offensive	0.00%	0.00%	0.00%	0.18%	0.03%
	porn	2.86%	0.33%	0.00%	1.63%	0.08%
	#langs =0% C	7	0	1	7	0
	#langs <50% C	**44**	**4**	**19**	**11**	**9**
	#langs >50% NL	13	0	0	7	1
	#langs >50% WL	1	0	0	3	4

Taken from [20]

High-Resource != High Quality





Taken from [20]

CCF's Internal Eval (2018) => We Use CLD2

	Languages	Accuracy				CPU Time
Test Set		dsl2014	europarl	twitter	tatoeba	cumul.
Docs		12600	21000	187461	19457	
MiB		3.3	3.4	18.8	0.9	
Languages		10	21	137	143	
cld2 [21]	160	.879	.989	.751	.792	4
cld2 (Java bind) [21, 22, 23]	83	.875	.986	.743	.606	13
cld2 (Java bind) [21, 22, 23]	**160**	**.880**	**.990**	**.758**	**.792**	**14**
cld2 Polyglot (Py bind) [24]	160	.879	.989	.748	.792	14
cld3 [25]	101	.830	.991	.595	.630	29
fasttext [26]	176	.880	.991	.753	.787	4
idNet [27]	463	.682	.997	.721	.680	3:07:07
langid (Python) [28]	97	.790	.992	.695	.604	19:19
langid (C) [29]	97	.790	.992	.695	.604	7
langid (Java) [30]	97	.790	.992	.695	.604	20
optimaize [31]	71	.740	.969	.534	.443	1:25
shuyo-cybozu [32]	52	.761	.993	.620	.420	56

About `CLD2`

- Two variants: 83 or 160 supported languages
- Detects up to 3 languages per document
- Plain text or HTML input, valid UTF-8
- Quadgram Naïve Bayesian classifier with support of
  - Writing system / Unicode script
  - Ignoring web-specific character sequences (*http*, *copyright*)
  - Top-level domain of URL
  - Language codes in HTML and HTTP metadata
  - Character encoding (`BIG5`, `KOI8-R`)
- C++ library, trained on a proprietary corpus; no support for re-training

Language Annotations in Common Crawl Data

- WARC metadata record


```
charset-detected: EUC-KR
languages-cld2: {
"reliable":true,
"text-bytes":2404,
"languages":[
{"code":"ko", "code-iso-639-3":"kor",
 "text-covered":0.97, "score":3735.0, "name":"Korean"},
{"code":"en", "code-iso-639-3":"eng",
 "text-covered":0.02, "score":1264.0, "name":"ENGLISH"}]}
```


- WAT (same as in WARC metadata record)
- WET

  `WARC-Identified-Content-Language: kor,eng`

- URL indexes: `kor,eng` (up to 3 language codes)

Improving Language Coverage and Balance

Two main goals of current efforts:

- **Expand coverage** of crawl for languages other than English (LOTE)
- **Improve language identification** for web: more languages, better fidelity

Initiative 1: The Web-Languages Project

- Crowd-sourced effort to improve crawling of low-resource languages
- Language, Community, Culture
- Show the crawler high-quality links to “steer” into the right direction
- Hopefully, this will influence the harmonic centrality ranks over time

(`https://github.com/commoncrawl/web-languages`)

The Web-Languages Project

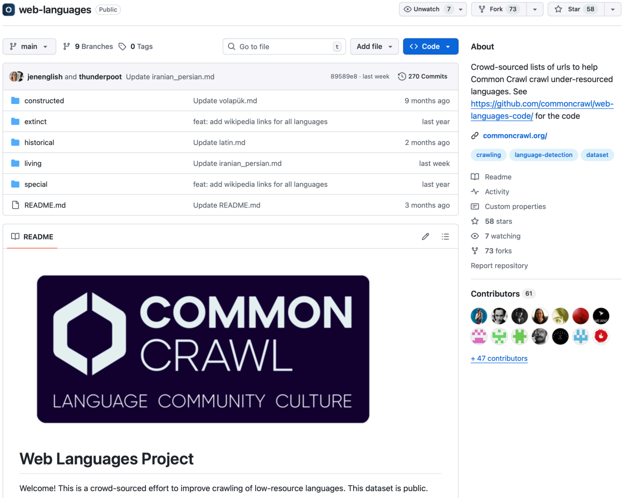
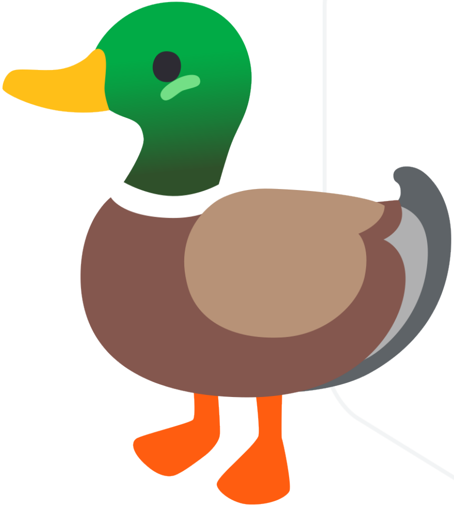


Figure 2: https://github.com/commoncrawl/web-languages

Initiative 2: Improving Language Identification for the Web



The LangID Project

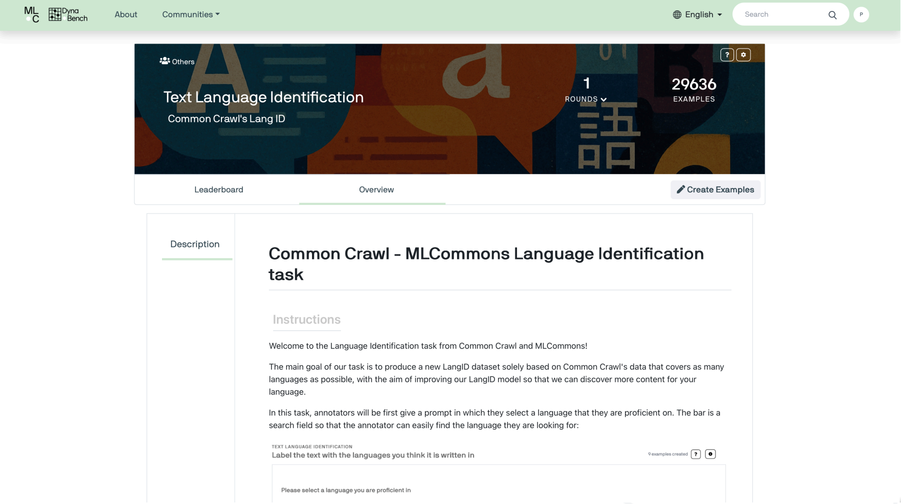


Figure 3: Now closed

Why LangID

- Existing models are old and unmaintained
- LangID models keep being used as filters but nobody studies the impact of this
- High coverage models remain largely under-evaluated due to lack of data
- The question of generalization across domains is under-studied
  - Most LangID datasets contain very clean data
- We want to expand the language coverage of Common Crawl and we need better signals
- LangID models need to accurately classify noisy web text
- Crowd-sourced data annotation task to collect manual annotations of web data

The Annotation Platform

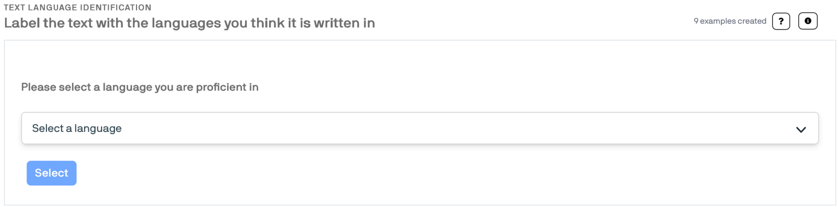


Figure 4: A screenshot of the annotation platform interface. The participant selects the language they want to annotate for.

The Annotation Platform

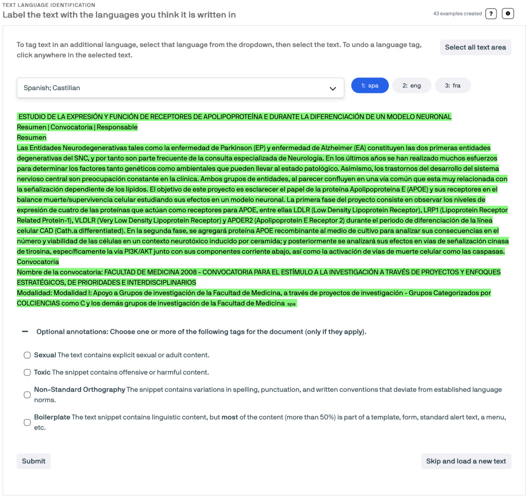


Figure 5: The participant has highlighted all text as Spanish.

The Annotation Platform

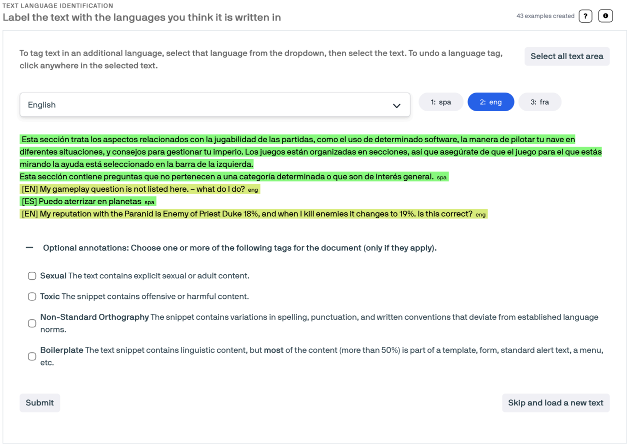


Figure 6: A screenshot of the annotation platform interface. The participant has highlighted the English and Spanish text in the extract in different colours.

The Annotation Platform

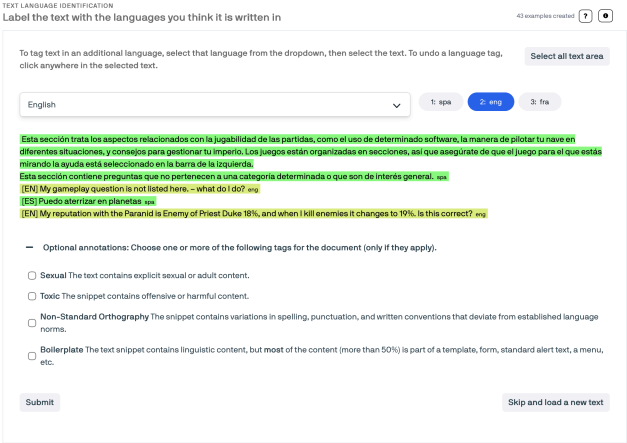


Figure 7: A screenshot of the annotation platform interface. The participant has highlighted the English and Spanish text in the extract in different colours.

The Incentives to Annotate

- Better LangID performance for your language!
- Contributors who annotated at least 100 documents are invited to be authors on the dataset paper
- Encourage the development of open-source LangID models
- More and higher-quality (and noisy!) multilingual data, especially for low-resource languages

The Organizations Behind the Effort

COMMON CRAWL

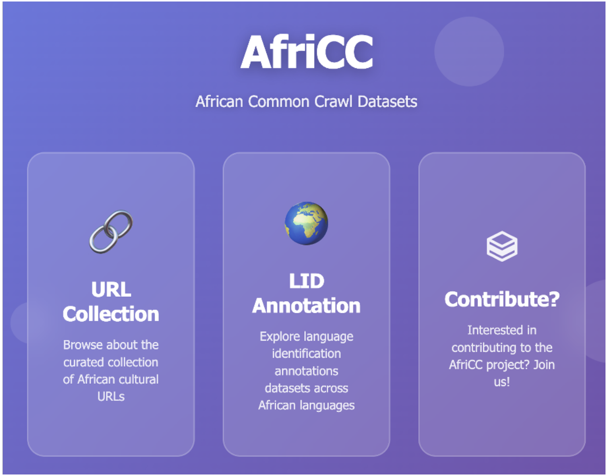
ML Commons

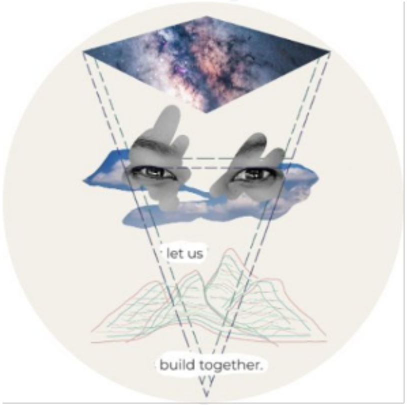
Factored

EleutherAI

JOHNS HOPKINS UNIVERSITY

Masakhane Hackathon





Masakhane

Masakhane Hackathon

AfriCC LID Instances per Language

Comprehensive visualization of African language representation in the dataset

63
African Languages

9,698
Total Entries

154
Average per Language

All African Languages

Search languages... | Sort A-Z | Sort by Region | Sort by Instances

Language ↑	Region	Instances
Acoli	East Africa	1
Aja (Benin)	West Africa	3
Amharic	Horn of Africa	574
Antankarana Malagasy	Malagasy	1
Bambara	West Africa	15

zoom.us

SEACrowd Hackathon

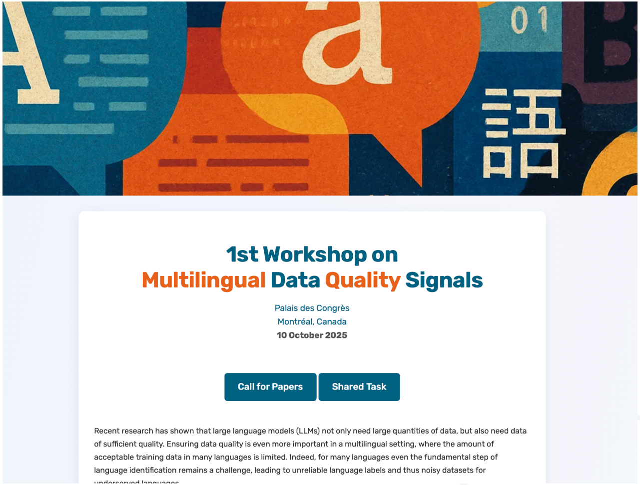
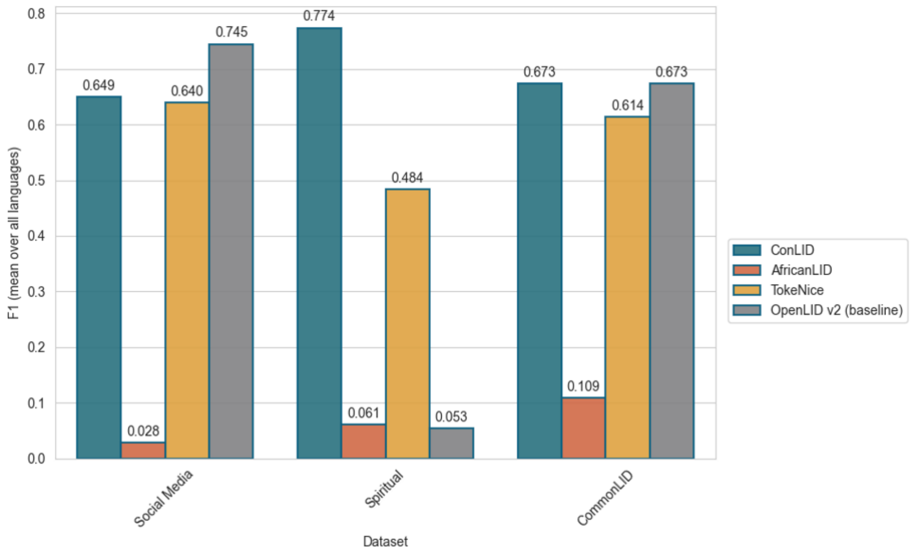


Figure 8: https://wmdqs.org

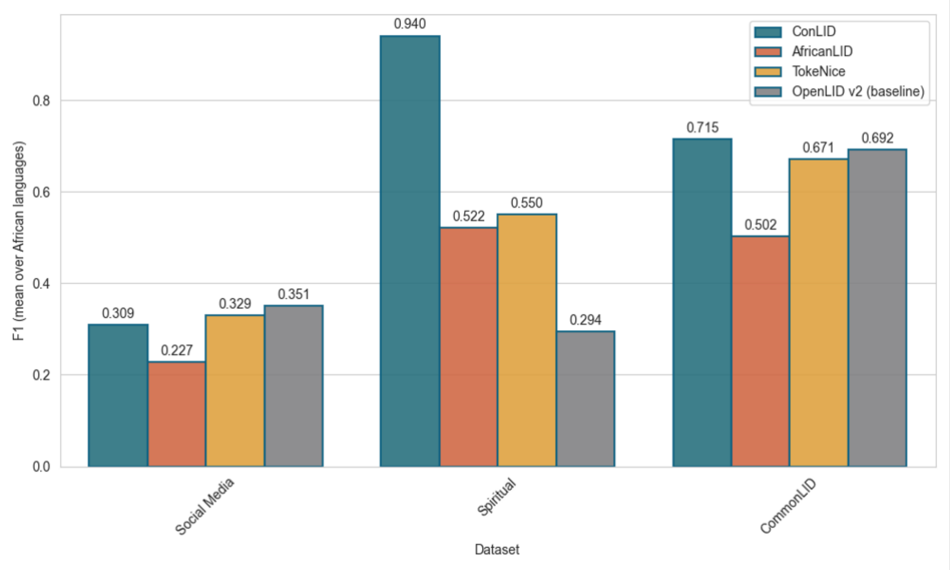
CommonLID v0.1 and a LangID Shared Taks

- CommonLID: 22K documents annotated or 200k annotated lines
  - 62 languages with >100 annotated lines
  - 42 languages with >500 annotated lines
  - 24 languages with >1000 annotated lines
  - 8 languages with >10000 annotated lines
- WMDQS LangID Shared Task
  - Evaluated on Dynabench
  - 18 different teams got into the shared task
  - 4 teams submitted models
  - 3 models ran on Dynabench
  - 3 Benchmarks were used: Social media dataset, spiritual text dataset and CommonLID

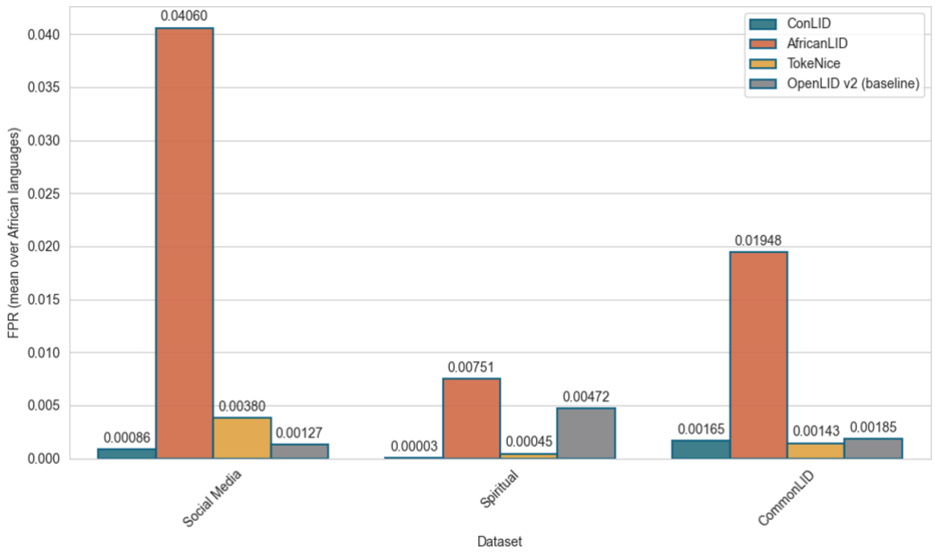
F1 Results: All Languages



F1 Results: African Languages



False-positive-rate: African Languages



Overview of CommonLID

Annotated data => LID evaluation dataset

- 373,230 lines
- 109 languages
- 78 languages with >100 lines
- 1–43,189 lines per class (mean 3424)
- 215.5 mean characters per line

All participants who annotated > 100 lines were offered co-authorship.

Geographic distribution of CommonLID

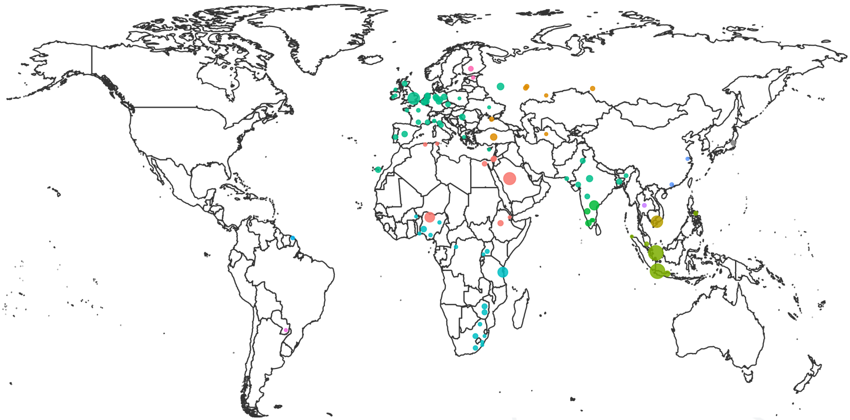


Figure 9: Illustrative geographical distribution of the language varieties in CommonLID. The dot size represents the number of annotated lines, and colour represents language family.

Creating CommonLID: Preprocessing

Converting to line-level annotations not trivial!

- Split on new lines
- Strip leading/trailing whitespace, de-duplicate line+language pairs
- Fuzzy match lines with multiple annotations

Creating CommonLID: Multiple Labels

Reconciling labels was a combination of automatic and manual processes

- Filter out short lines (< 10 characters)
- English contamination: `langid.py`, filter (most) lines $P(eng) > 0.7$
- Manual inspection of remaining multiple labels
  - Most remaining were clear mistakes => remove
  - Kept macro/micro language labels (both potentially valid)
  - Kept lines with multiple languages (none dominant)
- All languages audited (three spurious language classes removed)

Evaluating CommonLID

We evaluate 8 popular LID models using CommonLID + 5 evaluation datasets.

Fair comparison isn't easy:

- Label normalisation between eval sets and between model outputs (`iso639-lang`)
- Different models cover different languages: bigger not always better
- Web scale LID needs speed *and* fidelity!

Summary of Evaluated LID Models

Name	# langs.	Architecture	Data sources	Open data?
AfroLID [33]	517	Transformer	≈ 100M curated sents.	✗
CLD2 [21]	158	Naïve Bayes	Web pages (curated and scraped)	✗
fasttext [34]	218	FastText	"publicly available datasets"	✗
FUN-LangID [35]	1634	Common sub-strings	Web+Wikipedia+Bibles	✗
pyFranc [36]	414	Trigram distribution	UDHR	✓
gCLD3 [37]	99	Neural network	?	✗
GlotLID v3 [38]	1868	FastText	Curated open sources	✓
OpenLID-v2 [39]	193	FastText	Curated, audited open sources	✓

Table 1: Summary of LID models used.

Overlap in LID Model Coverage

	GlotLID	FUN-LangID	AfroLID	pyFranc	fasttext	OpenLID-v2	CLD2	gCLD3
GlotLID	**1868**							
FUN-LangID	1351	**1549**						
AfroLID	401	313	**515**					
pyFranc	362	312	93	**410**				
fasttext	202	180	49	166	**210**			
OpenLID-v2	190	157	47	156	180	**193**		
CLD2	128	153	30	121	114	104	**158**	
gCLD3	83	99	13	80	79	78	98	**99**

Table 2: Counts of mutual language coverage between LID models (models in descending order by coverage).

Macro-average F1 by model/evaluation set

	FLORES+ 209 languages		SmolSent 82 languages		UDHR-LID 418 languages		Bibles 1047 languages		Social Media 66 languages		CommonLID 109 languages	
	all	cov.	all	cov.	all	cov.	all	cov.	all	cov.	all	cov.
AL	15.9	67.9 (49)	42.3	69.4 (50)	15.2	67.6 (94)	12.1	69.8 (181)	1.5	**98.0** (1)	9.2	43.5 (23)
C2	45.0	87.8 (105)	40.6	83.2 (40)	24.0	81.7 (121)	4.4	73.8 (62)	76.8	85.9 (59)	49.5	**79.3** (68)
FT	80.3	92.2 (182)	48.1	83.8 (47)	27.0	68.2 (166)	3.3	43.8 (79)	**83.6**	84.9 (65)	49.3	72.6 (74)
FL	68.8	86.2 (165)	**75.9**	79.8 (78)	59.0	78.2 (314)	71.3	89.1 (837)	58.7	65.5 (58)	46.4	57.8 (86)
Fr	61.9	80.4 (160)	46.4	76.0 (50)	**93.4**	**95.9** (407)	7.6	52.1 (152)	62.4	66.4 (62)	39.3	61.3 (70)
C3	28.1	71.5 (78)	11.9	57.4 (17)	11.0	54.4 (80)	2.1	42.8 (47)	63.1	75.7 (54)	34.3	66.2 (55)
GL	**94.2**	**96.5** (204)	74.7	**91.4** (67)	73.7	84.4 (366)	**82.3**	**93.0** (927)	80.6	85.8 (62)	**60.4**	68.6 (96)
OL	83.5	90.4 (193)	45.4	82.7 (45)	25.1	67.3 (156)	3.3	44.6 (78)	83.4	88.7 (62)	47.4	68.0 (76)

Table 3: AfroLID (AL), CLD2 (C2), fasttext (FT), FUN-LangID (FL), pyFranc (Fr), CLD3 (C3), GlotLID (GL), OpenLID-v2 (OL). Scores over whole dataset (*all*) and on subset of covered languages (*cov.*). (x) = Count of languages in the eval. set covered by model.

Performance on Common Subsets of Languages

	High-coverage 1351 languages		Afro* 294 languages		Core 76 languages	
	F1 ↑	FPR ↓	F1 ↑	FPR ↓	F1 ↑	FPR ↓
GL	**87.8**	**0.006%**	**88.7**	**0.02%**	**91.6**	**0.1%**
FL	76.3	0.02%	73.5	0.08%	77.9	0.2%
AL	—	—	71.8	0.7%	—	—
Fr	—	—	—	—	77.1	0.4%
FT	—	—	—	—	77.7	0.5%
OL	—	—	—	—	78.5	0.5%
C2	—	—	—	—	84.7	0.2%
C3	—	—	—	—	61.6	1.4%

Table 4: Performance on combined test data. GlotLID (GL), FUN-LangID (FL), AfroLID (AL), pyFranc (Fr), fasttext (FT), OpenLID-v2 (OL), CLD2 (C2), CLD3 (C3).

Inference Speed Varies Greatly by Model

Model	Samples/s	Duration @$1e+6$ samples
CLD2	43735	22s
OpenLID-v2	13123	1m 16s
fasttext	10867	1m 32s
FUN-LangID	7237	2m 18s
GlotLID	3127	5m 19s
gCLD3	‡2026	8m 13s
Franc	520	32m 03s
AfroLID	†66	4h 12m 49s

Table 5: Inference speed for LID models on FLORES+. Inference done on a 14-core Apple M4 Pro chip with 64GB of RAM. †AfroLID uses transformers - performance is likely better on CUDA hardware. ‡Measured on an AMD EPYC 7351P (16 cores @ 2.4GHz) Linux machine with 256GB RAM as gCLD3 not usable on modern macOS.

Trade Off Between Inference Speed and F1

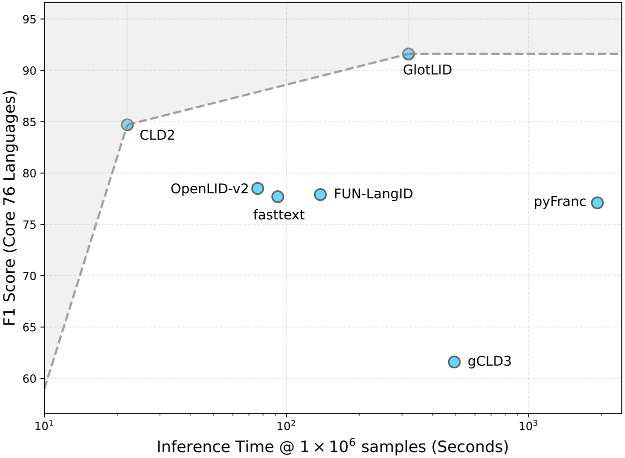


Figure 10: Inference speed vs. F1 (76 core languages, combined datasets)

GlotLID Beats LLM Performance for LID

	High-cov. 1351 langs.		Afro* 294 langs.		Core 76 langs.	
	F1 ↑	FPR ↓	F1 ↑	FPR ↓	F1 ↑	FPR ↓
GlotLID	90.0	0.01%	90.6	0.05%	93.5	0.13%
OpenAI GPT models						
4o-mini	48.5	0.09%	43.6	0.69%	81.0	0.54%
4o	62.0	0.07%	57.3	0.46%	89.0	0.33%
5-mini	60.4	0.06%	55.1	0.43%	76.4	0.53%
5	70.3	0.05%	66.6	0.29%	91.8	0.22%

Table 6: A comparison of large language models from OpenAI's GPT family (4o-mini, 4o, 5-mini, 5) and GlotLID (best performining LID model). We report performance on the combined but down-sampled test data (15k test samples), subsets as in .

ArXiv Paper

https://www.arxiv.org/abs/2601.18026

Summary

- Multilinguality is hard!
- Language Identification is hard!
  - Comparing and evaluating models is very difficult
  - Assessing language coverage is complicated, especially when different language registers are considered
  - We need more community initiatives to build better datasets
  - The Common Crawl Foundation will continue working on these language initiatives, but we need your help!
  - We need to develop better architectures and ensure long-term maintenance

Questions?

https://github.com/commoncrawl/presentations

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